K Nearest Neighbors

★ Concept of ML :-

Ex- Making Tea ⇝ Ingredients → Water, Tea leaves, Sugar, Milk

↓

Tea_1 = 2 Milk + 2 Water + 3 Tea Leaves + 4 Sugar

Tea_2 = 2 Milk + 1.5 Water + 2.5 Tea Leaves + 3.5 Sugar

The taste of both the Tea_1 & Tea_2 will be diff. because of the quantity of items used.

∴ The above eq. can also be written as -

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4$$

These are slopes or are also known as weights.

★ K-Nearest Neighbors :-

In Logistic Regression ⇝ Data is linear →

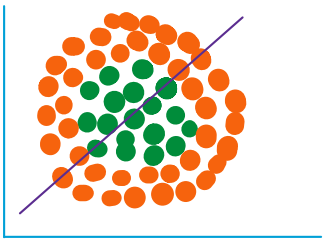


But, what if the data is non-linear

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● we will not be able to



- we will not be able to separate out the two classes perfectly using a line
- Drawing a curve to separate the two classes is not easy.

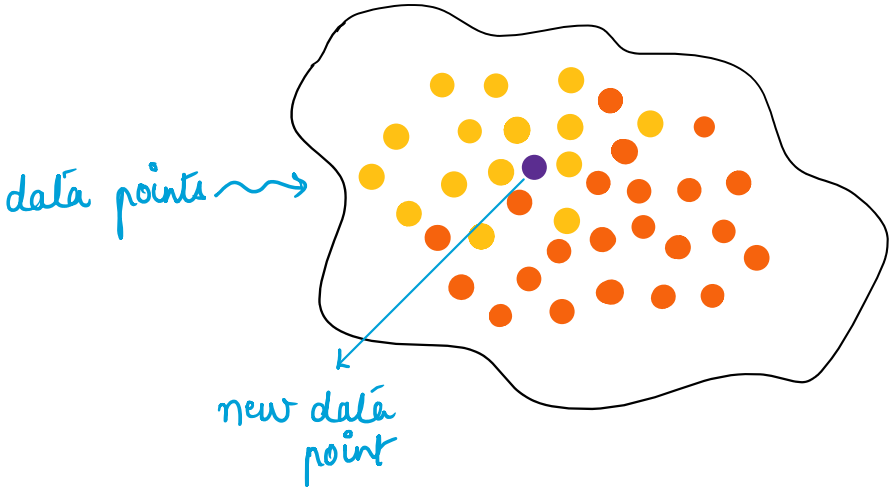
Solution is KNN

KNN is a supervised ML algorithm that do not focus on creating a line or a curve, etc. but it focuses on the neighboring values present around the data point.

* KNN for Classification :-
  - when the classes can be binary or can be Multiclass

Here KNN works with categorical data

Intuition behind KNN :-



point

Task → To find / predict the class of this new data point

Sol → We will look at the neighbors of new data point / query point

↓

Ques. Which Neighbour

Ans → Nearest Neighbors

Ques. How many Neighbors?

Ans → K-Nearest Neighbors

↳ K = constant

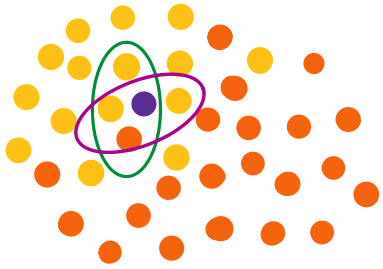
K = 1, 2, 3, 4, ----

Ques. How to select K-Nearest Neighbors?

Let K = 3

Ques. Which three?

Ans - Nearest three



Ques. How to find nearest 3?

Ans → For finding the nearest 3 neighbors, we can calculate the distance b/w the query point & other points

Ques. How to calculate the distance b/w data points?

Ans → Distance formula

↳ Euclidean Distance formula

$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} \quad \rightsquigarrow 2D$$

or

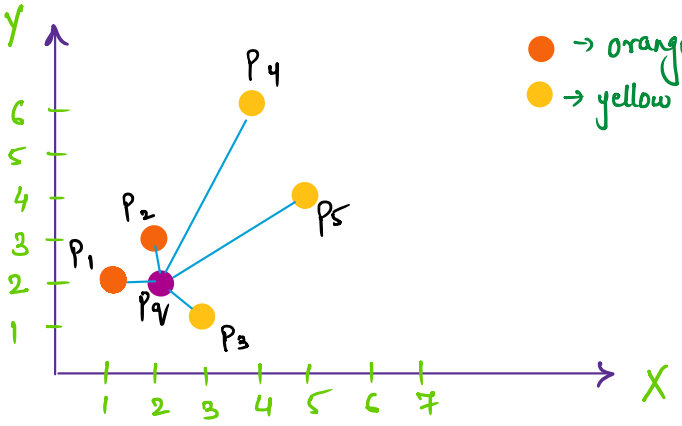
$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2 + \dots} \quad \rightsquigarrow nD$$

Here, $x, y, z \rightarrow$ belongs to dimension

$x_2, y_2, z_2 \rightarrow 1^{st}$ datapoint

$x_1, y_1, z_1 \rightarrow 2^{nd}$ datapoint

Ex-



→ orange class

→ yellow class

$P_1 = (1,2)$

$P_2 = (2,3)$

$P_3 = (3,1)$

$P_4 = (4,6)$

$P_5 = (5,4)$

P_q = query point = $(2,2)$

Calculate the distance b/w query point & all data points.

① for P_q & P_1

$P_q = (\overset{x_1}{2}, \overset{y_1}{2})$, $P_1 = (\overset{x_2}{1}, \overset{y_2}{2})$

$$d_1 = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

$$d_1 = \sqrt{(1-2)^2 + (2-2)^2}$$

$$d_1 = \sqrt{(-1)^2 + (0)^2}$$

$$d_1 = \sqrt{1+0} = 1$$

Similarly, we can calculate the other distances.

$$d_2 = 1, \quad d_3 = \sqrt{2}, \quad d_4 = \sqrt{20}, \quad d_5 = \sqrt{13}$$

$$d_3 = 1.414, \quad d_4 = 4.47, \quad d_5 = 3.6$$

After calculating the distances, we will select K

Let K=3

Here, nearest 3 distances are d_1, d_2 & d_3

So, 3 nearest neighbors to the query point are P_1, P_2, P_3

Ques. After finding the nearest neighbors what to do next?

Ans → find the class of the query point.

↓

How?

Go for majority voting to find the class of query point

In the above example → 2 points belongs to orange class
1 point belongs to yellow class

So, As per majority voting, the majority class is orange class.

∴ the query point will belong to orange class

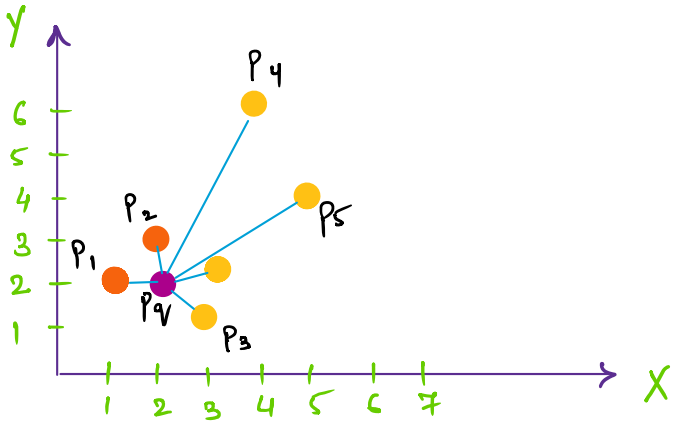
$P_q \rightarrow$ ●

This is how we calculate / find the prediction for any unknown data point using KNN [classification]

Ques. What if K=4 (even)

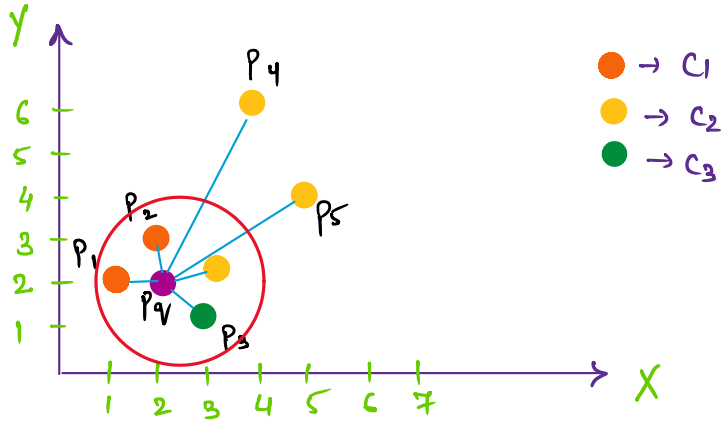
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Case I. It is difficult to find majority class in case of K (even) & binary classification



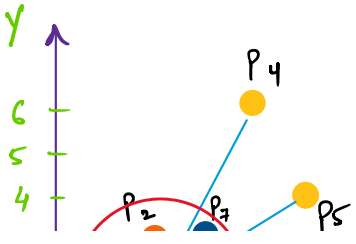
Case II. K = 4 (even) ; multiclass problem

Now, it is easier to select majority class

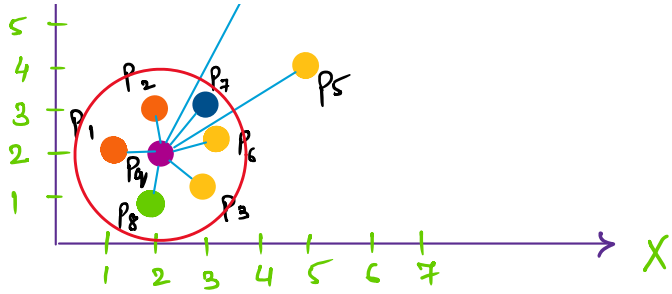


Case III. K=6 (even), multiclass prob.

Again, it is difficult to select majority

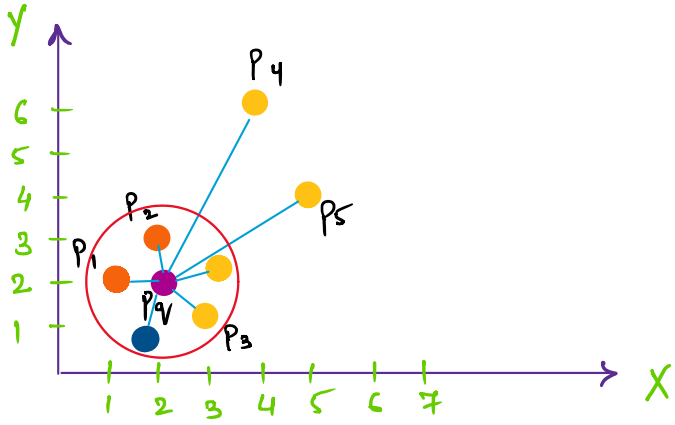


1 to select majority class



Case IV. K=5 (odd), multiclass

Again it will be difficult to select majority class



Final Ans → Always select K= odd.

Even after selecting K=odd, we get two or more majority classes, we can pick any class as the prediction from those majority classes.